Fee Query Assistant using Retrieval-Augmented Generation and Multilingual Tokenization: A Source-Grounded Bilingual Question-Answering System for College Fee Information

Darahasini[a)] and Venkata Sai[a)]

Student IDs: 2520030289 (Darahasini), 2520030278 (Venkata Sai) | Team-1

Instructor: Dr. Shaik Asif[a)]

[a)] Department of Computer Science and Engineering
Koneru Lakshmaiah Education Foundation, Bowrampet,
Hyderabad-500043, Telangana, India

Abstract

College students and parents routinely need accurate information about tuition fees, payment deadlines, late-fee rules, instalments, scholarship conditions and payment procedures. This information is scattered across notices, circulars and PDF documents, and it is usually written in a single language. General-purpose chatbots can answer fluently, but they may produce statements that no institutional document supports. This paper presents the design of a multilingual fee query assistant based on Retrieval-Augmented Generation (RAG). The system ingests approved, non-confidential fee documents, splits them into token-aware sections using a multilingual subword tokenizer, indexes the sections with multilingual embeddings in a vector database, retrieves the sections most relevant to a question asked in English or one selected local language, and generates a concise answer that cites its source. When the retrieved evidence is insufficient, the assistant states that the information was not found instead of guessing. The design follows the original RAG formulation and recent work on multilingual RAG, and we define an evaluation protocol covering answer correctness, retrieval relevance, citation coverage, language correctness and response time. This paper reports the problem analysis, literature review, architecture and evaluation plan completed at the first project review; implementation results will be reported once the prototype has been evaluated.

Keywords: *Retrieval-Augmented Generation, multilingual question answering, multilingual tokenization, fee information, document retrieval, student support.*

1 Introduction

Fee-related questions are among the most common and most time-sensitive queries that students and parents bring to a college's administrative office. Typical questions concern the amount payable in a given semester, the last date for payment without a penalty, the late-fee rule, whether fees can be paid in instalments, and what conditions apply to a scholarship or fee concession. The answers exist in official notices, circulars and PDF documents, but they are spread over many files, are revised from time to time, and are often written only in English. A parent who is more comfortable in a local language may therefore struggle to find or interpret a rule that directly affects a payment decision.

1.1 Problem Statement

Three issues motivate this work. First, fee information is scattered across notices and PDFs, so locating the right clause is slow. Second, students and parents need quick and reliable answers, but a language barrier makes official wording harder to understand. Third, general-purpose chatbots may produce confident answers that are not supported by any institutional document, which is a real risk when the answer concerns money and deadlines [10]. The problem addressed in this paper is how to give users fast, understandable, multilingual answers to fee questions while guaranteeing that every answer can be traced to an approved source document.

1.2 Objectives

- Answer common fee questions in English and one selected local language using only approved, non-confidential college documents.
- Display a source reference with every answer so that users can verify it against the official document.
- Decline to answer, with a clear “information not found” message, whenever the supplied documents do not contain sufficient evidence.
- Evaluate the prototype on answer correctness, retrieval relevance, citation coverage, language correctness and response time.

1.3 Contributions

This paper makes the following contributions at the proposal stage: (i) a problem analysis and societal-need argument for an institution-specific fee assistant; (ii) a literature review of RAG [1], multilingual RAG [2], dense multilingual retrieval and subword tokenization; (iii) a modular architecture for a source-grounded bilingual assistant, including a token-aware chunking step; (iv) three design rules derived from the primary references; and (v) a concrete, measurable evaluation plan.

The rest of this paper is organised as follows. Section 2 reviews related work. Section 3 presents the system architecture. Section 4 states the design rules drawn from the literature. Section 5 describes the evaluation methodology. Section 6 discusses feasibility and societal impact, Section 7 lists limitations and risks, Section 8 summarises project status and future work, and Section 9 concludes.

2 Related Work

2.1 Retrieval-Augmented Generation

Lewis et al. [1] introduced RAG as a way to combine a parametric language generator with a non-parametric memory of external documents. For each input, a retriever selects relevant passages and the generator conditions on them when producing the output. The approach was shown to be effective on knowledge-intensive tasks and has the practical advantage that the knowledge source can be updated without retraining the generator. For a fee assistant this matters directly: when a circular is revised, only the document index needs to be refreshed. RAG is therefore the core pipeline of our system, in which official fee-document sections are retrieved before any answer is generated.

2.2 Multilingual Retrieval-Augmented Generation

Ranaldi, Haddow and Birch [2] study RAG for multilingual knowledge-intensive question answering and compare multilingual retrieval strategies. Their work is relevant because a two-language assistant must handle a question in one language while the supporting evidence may be written in another. It motivates two checks that we adopt in our evaluation: whether the answer is produced in the language the user requested, and whether the retrieved content still supports the response after the language switch.

2.3 Dense Retrieval, Multilingual Embeddings and Tokenization

Dense passage retrieval [3] represents questions and passages as vectors and retrieves by similarity, which allows matching on meaning rather than exact keywords. Sentence-level encoders such as Sentence-BERT [4] make such embeddings efficient to compute, and multilingual encoders such as XLM-R [5] and multilingual E5 [6] map text from many languages into a shared space, so that a question in one language can retrieve a passage in another. Vector indexes such as FAISS [9] provide fast nearest-neighbour search. These models depend on subword tokenizers such as SentencePiece [7] and byte-pair encoding [8], which split words into smaller units so that rare words and morphologically rich languages can be represented. Because the same sentence can be split into a different number of subword tokens in different languages and scripts, chunk sizes and model input limits should be measured in tokens rather than characters. This is the role of multilingual tokenization in our design.

2.4 Hallucination and Grounded Answering

Surveys of natural language generation report that fluent models can produce content that is not supported by the source material [10]. Grounding the answer in retrieved text, showing the source, and abstaining when evidence is weak are the practical countermeasures that our design adopts.

2.5 Research Gap

The cited work addresses RAG and multilingual RAG in general settings. Table 1 contrasts common ways of answering fee questions with the proposed approach. The gap we address is an institution-specific, multilingual, source-grounded fee assistant that cites its source for every answer and declines when evidence is missing.

Aspect	Manual search of notices	General-purpose chatbot	Proposed fee assistant
Uses official documents	Yes	Not guaranteed	Yes, approved sources only
Local-language support	Rarely	Varies	English plus one local language
Shows source reference	Implicit	Usually no	Yes, with every answer
Admits missing information	Not applicable	Often answers anyway	Yes, “information not found”
Speed of finding an answer	Slow	Fast	Fast

Table 1. Qualitative comparison of approaches to answering fee queries.

3 Proposed System Architecture

3.1 Overview

The system has an offline indexing pipeline and an online query pipeline (Fig. 1). In the offline pipeline, approved fee documents are converted to text, split into sections, embedded and stored in a vector database together with metadata such as document title, date and page. In the online pipeline, a user question is embedded, the most relevant sections are retrieved, an evidence check decides whether the sections are sufficient, and a language model generates a concise answer with a source reference. If the evidence is insufficient, the system returns an “information not found” message.

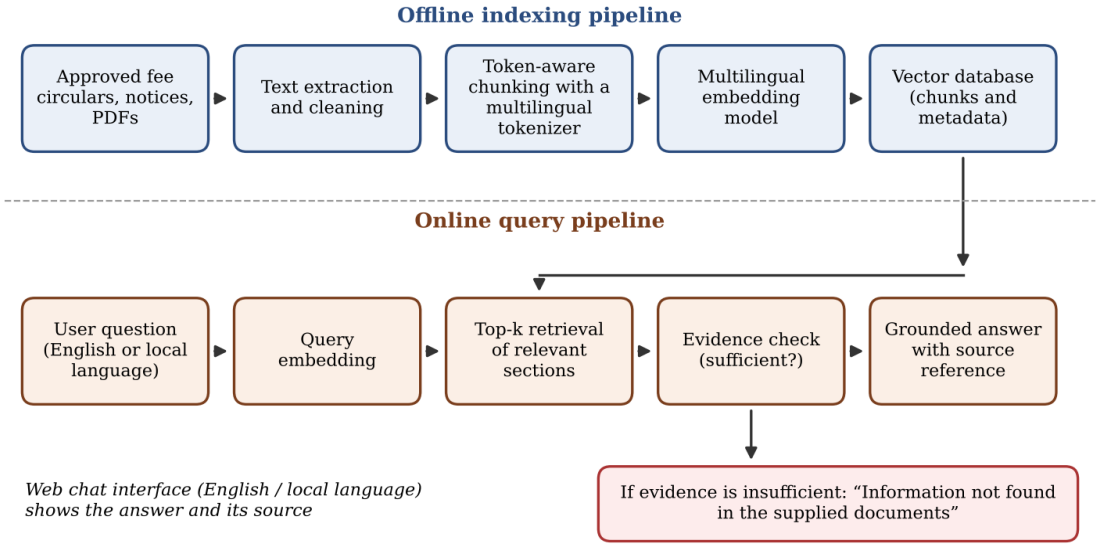


Fig. 1. Architecture of the Fee Query Assistant: offline indexing pipeline and online query pipeline.

3.2 Document Ingestion and Text Extraction

The input is limited to approved, non-confidential fee circulars and notices. Text is extracted from each PDF and cleaned of headers, footers and page artefacts. Tables of fee amounts are kept together with their headings so that a figure is never separated from the course, year or category it belongs to. Scanned documents will require optical character recognition and are handled as a later extension.

3.3 Token-Aware Chunking with a Multilingual Tokenizer

Each document is split into sections of limited size with a small overlap so that a rule is not cut in the middle of a sentence. Section length is measured in tokens produced by the same subword tokenizer used by the embedding model, so that no chunk exceeds the model’s input limit in either language. Metadata (document name, section heading, page) is stored with every chunk and later supplies the source reference.

3.4 Multilingual Embeddings and Vector Database

Chunks and queries are encoded with a multilingual embedding model so that a question in the local language can retrieve an English section and vice versa [5], [6]. The vectors are stored in a vector database that supports similarity search and metadata filtering [9].

3.5 Retrieval and Evidence Check

For each question the top-k most similar sections are retrieved. An evidence check then compares the retrieval scores with a threshold and verifies that at least one retrieved section is relevant to the question. If the check fails, generation is skipped and the assistant replies that the information is not available in the supplied documents. The threshold and the value of k will be tuned on a development set of questions.

3.6 Grounded Answer Generation and Citation

A language model receives the question and only the retrieved sections, with an instruction to answer in the user's language, to use nothing outside the sections, and to name the source of each statement. The reply shows the answer together with the document name and section or page reference, so the user can check it against the official file.

3.7 User Interface and Technology Components

A simple web chat interface lets the user select English or the local language, type a question and read the answer with its source. Table 2 summarises the components and their roles.

Component	Role in the system
Text extraction	Convert approved PDFs and notices to clean text with page information
Multilingual tokenizer	Measure and split text into subword tokens for token-aware chunking
Multilingual embedding model	Map questions and sections in both languages into a shared vector space
Vector database	Store chunk vectors and metadata; return top-k similar sections
Language model	Generate a concise answer in the requested language from retrieved sections only
Web chat interface	Accept questions, show answers, source references and the "not found" message

Table 2. System components and their roles.

4 Design Rules Derived from the Literature

The two primary references shape the design as follows. Paper [1] provides the architecture: retrieve evidence first, then generate the answer. Paper [2] provides guidance for multilingual use: test whether the answer is in the requested language and whether the retrieved content still supports the response. These findings lead to three project rules:

- **Rule 1 – Approved sources only.** The assistant answers only from approved, non-confidential fee documents.
- **Rule 2 – Visible sources.** Every answer displays its source reference.
- **Rule 3 – Decline when unsure.** If the evidence is insufficient, the assistant says so and does not guess.

5 Evaluation Methodology

5.1 Test Set

We will prepare a set of representative fee questions that covers tuition amounts, payment deadlines, late-fee rules, instalments, scholarship conditions and payment procedures. Each question will be written in both English and the local language, with a reference answer and the reference source section. The set will also contain questions whose answers are not present in the documents, to test whether the assistant correctly declines. The size of the set will be fixed when the document collection is finalised.

5.2 Metrics

Metric	Definition	How measured
Answer correctness	Answer states the same facts (amounts, dates, conditions) as the reference answer	Manual grading: correct, partly correct, incorrect
Retrieval relevance	Retrieved sections include the reference source section	Hit@k and mean reciprocal rank over the test set
Citation coverage	Share of answers that display a source, and share of those sources that are correct	Count and manual check
Language correctness	Answer is written in the language requested by the user	Manual check or language identification
Response time	Time from question submission to the complete answer	Mean and 95th percentile latency
Abstention accuracy	Unanswerable questions are declined, answerable ones are not	Confusion counts on the test set

Table 3. Planned evaluation metrics.

5.3 Comparison and Ablation

To show the value of each design decision, the system will be compared with (a) a language model answering without retrieval and (b) keyword search over the same documents. Ablations will vary the number of retrieved sections k, the chunk size, the evidence threshold, and the question language, so that language correctness and source consistency can be reported separately for English and the local language, as suggested by [2].

6 Feasibility and Societal Impact

The prototype is feasible with existing components: document text extraction, multilingual embeddings, a vector database, a language model and a simple web interface. The initial scope is limited to common fee questions and two languages, which keeps the document collection and the test set manageable. Socially, the assistant makes official information more accessible to students and parents, supports users who prefer a local language, and can reduce the number of routine fee-related queries that reach administrative staff.

7 Limitations and Risks

- **Document currency.** Answers are only as current as the indexed documents; superseded circulars must be removed or dated.
- **Scope.** The assistant answers general fee-policy questions from public documents. It does not access individual student accounts or confidential records.
- **Extraction quality.** Tables and scanned PDFs can be extracted incorrectly, which may affect retrieval and amounts quoted.
- **Language quality.** Answer fluency and accuracy in the local language may be lower than in English and will be measured separately.
- **Residual errors.** Grounding reduces but does not eliminate unsupported statements [10]; the source reference lets users verify an answer, and official office confirmation remains the final authority.
- **Proposal stage.** This paper does not yet report experimental results.

8 Project Status and Future Work

At the first review the abstract, literature survey, research-paper references and repository structure are ready on GitHub. The next steps are to collect and approve the fee documents, select the embedding model, vector database and language model, implement the indexing and query pipelines, build the web interface, create the bilingual test set, and run the evaluation in Section 5. Possible extensions include optical character recognition for scanned notices, support for more languages, and a feedback option for users to flag incorrect answers.

9 Conclusion

This paper proposed a multilingual, source-grounded fee query assistant that combines Retrieval-Augmented Generation with token-aware multilingual indexing. By answering only from approved documents, showing a

source for every answer and declining when evidence is missing, the design aims to give students and parents fast, understandable and verifiable fee information while reducing routine administrative queries. The evaluation plan defines how correctness, retrieval quality, citation coverage, language correctness and response time will be measured once the prototype is built.

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